

Infosys Springboard Virtual Internship 6.0 Completion Report

Team Details

Batch Number: 13

Start date: 02 Feb 2026

Names:

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Internship Duration: 8 Weeks

1. Project Title

HOTELREVAI - AI DRIVEN REVENUE ANALYSIS FOR HOTELS

2. Project Objective

The main objective of the HotelRevAI project is to analyse hotel booking data and generate meaningful insights using data analytics and AI-driven visualization techniques.

The project aims to:

- Transform raw hotel booking data into structured and meaningful insights
- Build an efficient data model for analysis
- Create key business KPIs using DAX
- Develop an interactive dashboard for decision-making

This project helps hotel managers to optimize revenue, understand customer behaviour, and improve business performance.

3. Project description in detail

The HotelRevAI project is a complete data analytics solution that focuses on analysing hotel booking data from 2015 to 2017 to extract valuable business insights.

Approach

The project was executed in multiple stages:

- Data collection and understanding
- Data cleaning and preprocessing
- Feature selection and data modelling
- KPI creation using DAX
- Dashboard development and visualization
- AI-driven insights generation

Technologies Used

- **Power BI** – Dashboard creation and visualization
- **DAX (Data Analysis Expressions)** – KPI and metric calculations
- **CSV Dataset** – Hotel booking dataset
- **Data Modelling Techniques** – Relationship building and optimization

Key Features

- Analysis of booking patterns and customer behaviour
- Revenue trend analysis (monthly and yearly)
- Market segment performance analysis
- Seasonal demand identification
- AI-powered insights and visual dashboards

Real-World Impact

This project helps hotels:

- Optimize pricing strategies
- Improve demand forecasting
- Understand customer segments
- Increase operational efficiency
- Make data-driven decisions

4. Timeline Overview

Week	Activities Planned	Activities Completed
Week 1	Project orientation and dataset understanding	Understood project scope and explored dataset
Week 2	Data loading and preprocessing	Imported dataset and checked data quality
Week 3	Data cleaning	Removed missing values, duplicates, corrected data
Week 4	Feature selection and modelling	Selected key features and created relationships
Week 5	DAX implementation	Created KPIs like revenue, ADR, bookings
Week 6	Dashboard development	Built initial dashboard visuals
Week 7	Dashboard enhancement	Added slicers, filters, and interactivity
Week 8	Final review and AI insights	Finalized dashboard and extracted insights

5a. Key Milestones

Milestone	Description	Date Achieved
Project Kick off	Orientation and problem understanding	05 Feb 2026
Prototype / First Draft	Initial dashboard creation	11 Feb 2026
Mid-Term Review	Evaluation of progress	20 Feb 2026
Final Submission	Completed dashboard and report	13 Mar 2026
Presentation	Final project presentation	24 Mar 2026

5b. Project execution details

The project execution followed a structured and milestone-based approach.

Milestone 1: Orientation and Data Loading

This initial phase focused on understanding the project requirements and preparing the dataset for analysis.

Orientation Phase

- Understood the business problem of hotel revenue optimization
- Identified key analysis areas such as bookings, revenue trends, and customer segmentation
- Explored the dataset structure including columns like booking dates, ADR, market segment, and customer type

Data Loading

- Imported the hotel booking dataset (CSV format) into Power BI
- Verified column data types (date, numeric, categorical)
- Ensured dataset completeness and consistency

Data Cleaning & Preprocessing

- Removed missing or null values to avoid incorrect analysis
- Eliminated duplicate records to maintain data integrity
- Standardized column names and categorical values
- Corrected inconsistent data formats
- Handled outliers in numerical fields such as ADR

Feature Engineering

- Created new calculated fields such as:
 - Total Stay Duration (weekend + weekday nights)
 - Revenue estimation using ADR
- Prepared the dataset for further modelling and analysis

Outcome: A clean, structured dataset ready for analysis and modelling.

Milestone 2: Data Loading & Modelling

This phase focused on structuring the dataset for efficient and scalable analysis.

Feature Selection

Important attributes selected include:

- Booking and arrival dates
- Room type classification
- Guest details and composition
- Average Daily Rate (ADR)
- Lead time (booking gap)
- Cancellation status
- Market segment and customer type

These features were chosen to support meaningful business insights.

Data Modelling

- Built relationships between tables using keys
- Developed a **date hierarchy** (Year - Month - Day) for time-based analysis
- Structured the dataset into a logical model to improve performance

Optimization

- Removed unnecessary columns to reduce model size
- Ensured efficient relationships to improve dashboard performance

Outcome: A well-structured and optimized data model enabling faster and accurate analysis.

Milestone 3: DAX & Business Metrics

This phase focused on creating business metrics using DAX (Data Analysis Expressions).

Key Measures Created

- Total Bookings - Count of all confirmed bookings
- Total Revenue - Estimated revenue based on ADR and stay duration
- Average Stay Duration - Average number of days guests stay
- Average Lead Time - Average time between booking and arrival
- Average Daily Rate (ADR) - Average room price per night
- Cancellation Rate - Percentage of cancelled bookings

Business Metrics & Insights

- Identified revenue trends across months and years
- Analysed seasonal demand patterns
- Evaluated performance of different customer segments
- Studied booking channels contribution

Outcome: Strong analytical foundation with meaningful KPIs for decision-making.

Milestone 4: Visualization, Storytelling and AI

This phase focused on transforming data into actionable insights using dashboards and visualizations.

Dashboard Development

- Created an interactive Power BI dashboard
- Designed KPI cards:
 - Total Revenue
 - Total Bookings
 - Average Revenue

Visualizations

- Revenue trends (year-wise)
- Bookings by hotel type
- Customer distribution analysis
- Market segment performance

Interactivity Features

- Year slicer (2015–2017)
- Filters for hotel type and customer segment
- Cross-visual interactions

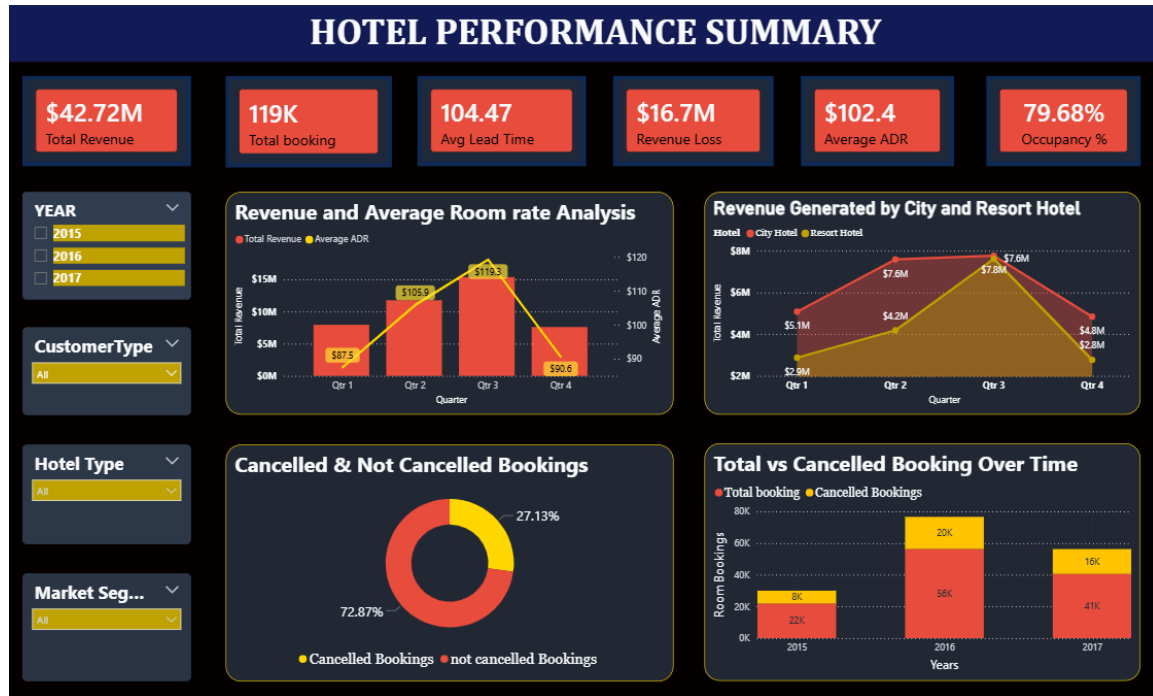
AI-Driven Insights

- Converted raw data into meaningful insights
- Enabled quick understanding of trends and patterns
- Supported data-driven decision-making

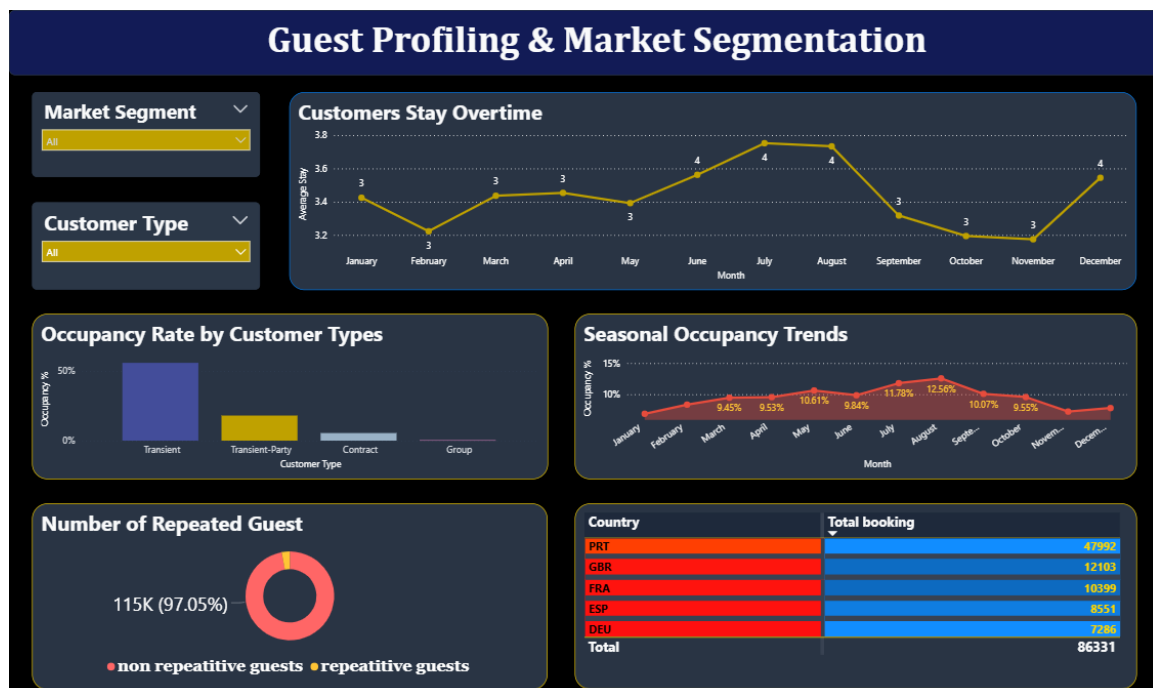
Outcome: A fully functional, interactive dashboard enabling real-time business insights.

6. Snapshots / Screenshots

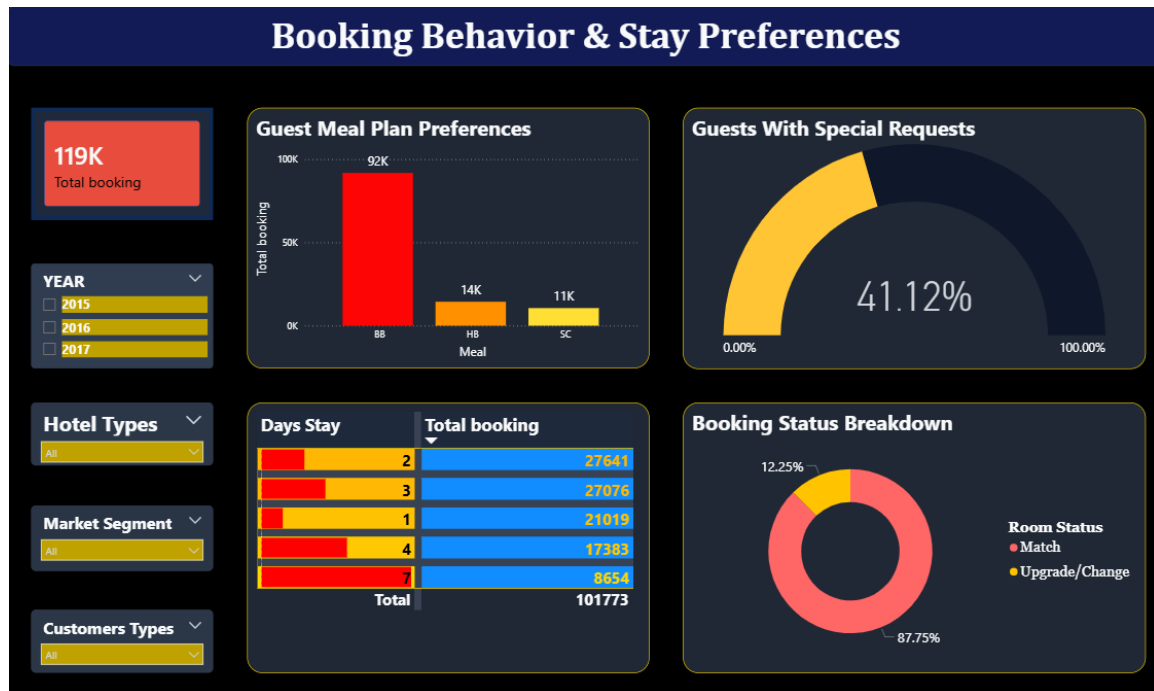
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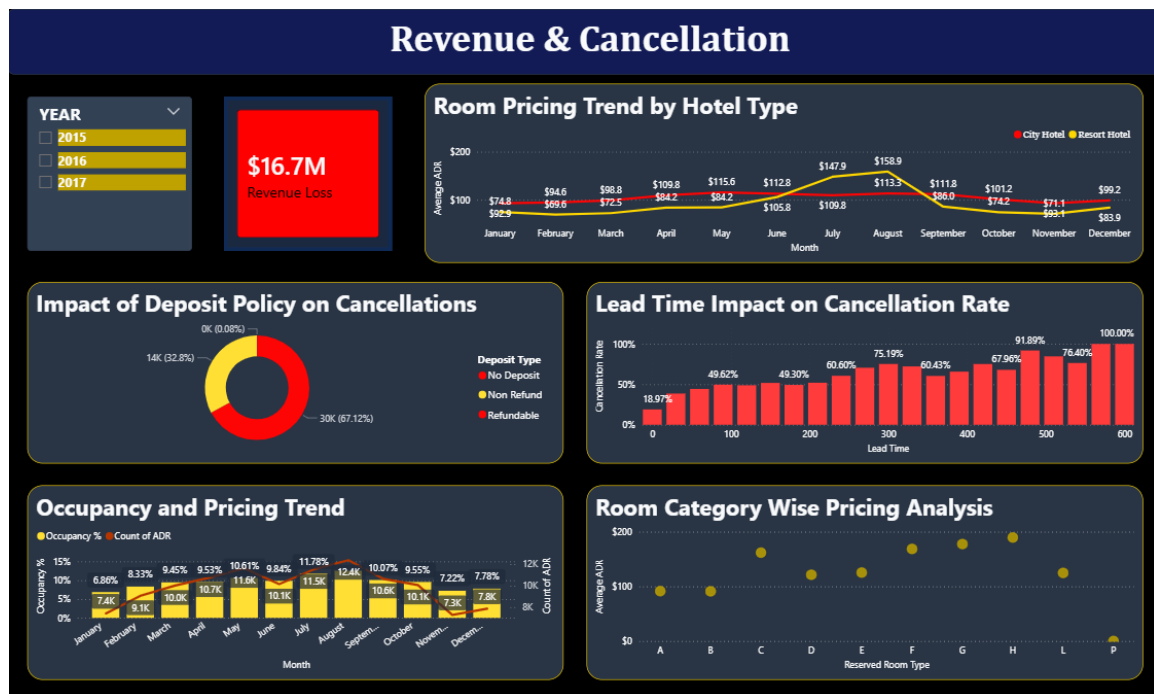
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7. Challenges Faced

Data Quality Issues

- The dataset contained missing, inconsistent, and sometimes incorrect values, which affected the accuracy of analysis.
- **Solution:** These issues were resolved using Power Query by cleaning the data, removing null values, and standardizing formats.

Complex Data Modelling

- Understanding relationships between multiple attributes and creating an optimized data model was challenging.
- **Solution:** This was overcome by carefully analysing the dataset structure and applying proper relationship-building techniques.

Learning DAX Formulas

- Writing accurate DAX measures for KPIs required practice and a clear understanding of logic.
- **Solution:** Improved through continuous experimentation, testing, and debugging of formulas.

Dashboard Design and Visualization

- Choosing the right visualizations and maintaining a clean, user-friendly dashboard layout was difficult.
- **Solution:** Addressed by iterating designs, testing different chart types, and focusing on clarity and usability.

Time Management within the Team

- Coordinating tasks and managing time effectively among team members was sometimes challenging, especially when balancing different schedules.
- **Solution:** Improved by dividing work clearly and maintaining regular progress updates.

Team Communication and Idea Discussions

- At times, communicating ideas and aligning decisions within the team required effort, especially during brainstorming and feedback sessions.
- **Solution:** This was managed by maintaining open discussions, actively listening to each other, and encouraging feedback to improve the project outcomes.

8. Learnings & Skills Acquired

Technical Skills

- Power BI dashboard development
- DAX formulas and KPI creation
- Data preprocessing and transformation
- Data modelling and relationships

Analytical Skills

- Identifying trends and patterns
- Business insight generation
- Data-driven decision making

Soft Skills

- Problem-solving
- Time management
- Analytical thinking

9. Testimonials from team

The HotelRevAI project was a highly enriching and collaborative learning experience for the entire team. It provided an opportunity to work on a real-world dataset and apply data analytics concepts to solve practical business problems in the hospitality domain. Throughout the internship, we actively shared ideas, discussed different approaches, and supported each other in overcoming technical challenges. This collaborative environment helped us improve not only our technical skills but also our communication and teamwork abilities.

Working on this project enhanced our understanding of Power BI, DAX, and data modelling, while also strengthening our analytical thinking and problem-solving skills. The process of converting raw data into meaningful insights gave us valuable exposure to real-world business intelligence practices. Overall, this experience boosted our confidence in handling data-driven projects and prepared us for future roles in the field of data analytics and business intelligence.

10. Conclusion

The HotelRevAI project successfully demonstrates the effective use of data analytics and visualization techniques to transform raw hotel booking data into meaningful business insights. By leveraging tools like Power BI and DAX, the project provided a structured approach to analyzing key factors such as revenue trends, customer behavior, and booking patterns.

The development of an interactive dashboard enabled clear visualization of important KPIs, allowing users to easily interpret data and make informed decisions. The integration of analytical methods and AI-driven insights helped in identifying seasonal trends, customer preferences, and revenue opportunities.

Overall, this internship project enhanced both technical and analytical skills while providing practical exposure to real-world business intelligence applications. The knowledge and experience gained through this project will be valuable for future work in data analytics and related domains.

11. Acknowledgements

I would like to express my sincere gratitude to **Infosys Springboard** for providing this valuable internship opportunity, which allowed me to gain practical exposure to real-world data analytics and business intelligence. I am deeply thankful to my mentors for their continuous guidance, support, and encouragement throughout the project. Their valuable insights and feedback played a crucial role in successfully completing this internship. I would also like to thank our wonderful colleagues and friends for listening to our ideas, asking thoughtful questions, and providing valuable feedback and suggestions that helped improve our project.

Finally, I extend my gratitude to everyone who contributed directly or indirectly to the successful completion of this project.